Question 1 — Pairwise Correlation with Missing Data

1(a) Pairwise correlation matrix

For each pair of series, I retained every row on which both members of the pair were observed and calculated the Pearson correlation on that pairwise-complete sample. Because the missing-value pattern differs across pairs, the effective sample is not identical for every entry.

	x1	x2	x3	x4	x5
x1	1.000000	0.332836	0.418462	0.255285	0.732272
x2	0.332836	1.000000	0.251035	0.287709	0.787538
x3	0.418462	0.251035	1.000000	0.483580	0.684257
x4	0.255285	0.287709	0.483580	1.000000	0.782632
x5	0.732272	0.787538	0.684257	0.782632	1.000000

1(b) Variances using all available observations

I calculated each sample variance independently, deleting missing values only from the series being measured. The denominator is $n_i - 1$. This preserves every valid observation in each individual series.

Series	Valid observations	Sample variance
x1	41	0.0004060739
x2	41	0.0004800663
x3	36	0.0001511102
x4	41	0.0008417388
x5	42	0.0002731496

1(c) Positive semi-definite test

A real symmetric matrix is positive semi-definite if and only if every eigenvalue is nonnegative. The eigenvalues of the pairwise correlation matrix are: [−0.08927867, 0.46133980, 0.71908517, 0.82343722, 3.08541647]

The matrix is not positive semi-definite because its smallest eigenvalue is −0.08927867. This is a genuine negative value rather than numerical rounding. Pairwise deletion can produce this result because different correlations are estimated from different subsets of dates, so the entries need not form one internally consistent joint correlation structure.

1(d) Higham nearest positive semi-definite correlation matrix

I applied Higham's alternating-projection algorithm. Each iteration projects onto the positive semi-definite cone by replacing negative eigenvalues with zero, then projects onto the unit-diagonal set by restoring every diagonal entry to one.

	x1	x2	x3	x4	x5
x1	1.000000	0.349452	0.425942	0.270782	0.694572
x2	0.349452	1.000000	0.260599	0.307523	0.739337
x3	0.425942	0.260599	1.000000	0.492500	0.662558
x4	0.270782	0.307523	0.492500	1.000000	0.737677
x5	0.694572	0.739337	0.662558	0.737677	1.000000

The corrected matrix has eigenvalues: $[-1.62\times10^{-12}, 0.45008153, 0.70347611, 0.80670847, 3.03973389]$

The first eigenvalue is zero to numerical precision; the displayed -1.62×10^{-12} is floating-point error. Therefore, the corrected matrix is positive semi-definite.

Entry	Original	Higham	Absolute change
(x2, x5)	0.787538	0.739337	0.048201
(x4, x5)	0.782632	0.737677	0.044955
(x1, x5)	0.732272	0.694572	0.037700
(x3, x5)	0.684257	0.662558	0.021699
(x2, x4)	0.287709	0.307523	0.019814

The entries involving x5 changed the most. The largest change was the correlation between x2 and x5, followed by the correlations between x4 and x5 and between x1 and x5.

1(e) Covariance matrix

Let D be the diagonal matrix of full-sample standard deviations from part (b), and let R^H be the Higham correlation matrix from part (d). I formed the covariance matrix as $\Sigma = D\, R^H\, D$. Consequently, each off-diagonal covariance is $R^H_{ij}\sigma_i\sigma_j$, while the diagonal exactly reproduces the variances from part (b).

	x1	x2	x3	x4	x5
x1	0.0004060739	0.0001542911	0.0001055115	0.0001583111	0.0002313238
x2	0.0001542911	0.0004800663	0.0000701891	0.0001954867	0.0002677277
x3	0.0001055115	0.0000701891	0.0001511102	0.0001756475	0.0001346081
x4	0.0001583111	0.0001954867	0.0001756475	0.0008417388	0.0003537164
x5	0.0002313238	0.0002677277	0.0001346081	0.0003537164	0.0002731496

Question 2 — VaR and ES for a Single Stock

2(a) VaR at 95% and 99%

I first converted the 501 prices to 500 arithmetic daily returns. I estimated the market model by ordinary least squares. The fitted coefficients and distribution inputs are:

Quantity	Estimate
α	−0.00030402
β	1.19331076
SPY mean	0.00025477
SPY sample standard deviation	0.01097207
Residual sample standard deviation	0.01459005

Because rSPY and ε are assumed to be independent normal variables, their linear combination is also normal. The implied XYZ mean and standard deviation are:

$$\mu XYZ = \alpha + \beta\mu SPY = 0.00000000$$

$$\sigma XYZ = \sqrt{(\beta^2\sigma^2 SPY + \sigma^2\varepsilon)} = 0.01960353$$

For confidence level c, I used the lower return quantile and reported the loss as a positive number: $VaR^c = -[\mu XYZ + \sigma XYZ\ \Phi^{-1}(1-c)]$. The number of shares affects dollar VaR but cancels when the result is expressed as a percentage of position value.

Confidence level	Return cutoff	VaR
95%	−3.2245%	3.2245%
99%	−4.5605%	4.5605%

2(b) ES at 95% and 99%

Expected shortfall is the average loss conditional on the return falling below the corresponding VaR cutoff. For a normal return, with $p = 1 - c$ and $z_p = \Phi^{-1}(p)$, I used:

$$ES^c = -\mu XYZ + \sigma XYZ\ \varphi(z_p)/p$$

Confidence level	VaR	Expected shortfall
95%	3.2245%	4.0436%
99%	4.5605%	5.2248%

ES is larger than VaR at both confidence levels because it averages losses beyond the quantile rather than reporting only the boundary of the tail. The increase from 95% to 99% also reflects the more extreme loss region being measured.

Question 3 — Copula VaR and ES for a Two-Book Portfolio

3(a) Marginal distribution fitting and selection

I converted the 1,001 prices into 1,000 arithmetic daily returns for each stock. Each Normal, Student's t, and normal inverse Gaussian (NIG) distribution was fitted by maximum likelihood. I compared models with the small-sample corrected Akaike information criterion:

$$\mathrm{AICc} = -2\ell + 2\mathrm{k} + 2\mathrm{k}(\mathrm{k}+1)/(\mathrm{n}-\mathrm{k}-1)$$

Here k equals 2 for the Normal, 3 for Student's t, and 4 for NIG. Lower AICc is preferred because it rewards likelihood while charging more flexible distributions for their extra parameters.

Normal distribution fits:

Stock	μ	σ	AICc
S1	0.00090370	0.01182025	−6033.994
S2	0.00080026	0.01533991	−5512.706
S3	0.00042361	0.01888852	−5096.513
S4	0.00077999	0.00992765	−6382.974
S5	0.00070877	0.01894165	−5090.895
S6	0.00080469	0.01507758	−5547.203
S7	0.00069273	0.01366672	−5743.694
S8	0.00037400	0.01710306	−5295.107
S9	0.00174602	0.02063642	−4919.506
S10	0.00016480	0.01305470	−5835.325

Student's t distribution fits:

Stock	ν	Location	Scale	AICc
S1	355368542.589	0.00090366	0.01182027	−6031.982
S2	4.857877	0.00058139	0.01167261	−5627.360
S3	4.160482	0.00229110	0.01365721	−5238.463
S4	339036291.501	0.00078000	0.00992769	−6380.962
S5	6.289248	0.00083326	0.01557655	−5150.869
S6	3.823756	0.00244489	0.01065489	−5688.740
S7	319103.363	0.00069242	0.01366808	−5741.682
S8	3.483289	0.00065411	0.01142390	−5493.288
S9	4.385214	0.00338341	0.01531014	−5036.762

Stock	ν	Location	Scale	AICc
S10	4.121270	0.00040472	0.00942699	−5974.857

NIG distribution fits:

Stock	α	β	δ	μ	AICc
S1	2086.485600	−588.204192	0.25747326	0.07655696	−6030.215
S2	73.206598	4.863823	0.01659306	−0.00030457	−5622.063
S3	79.836689	−36.344624	0.01965684	0.01047387	−5281.641
S4	3368.189374	−984.957946	0.29028696	0.08954907	−6379.120
S5	73.576752	−2.232562	0.02595840	0.00149685	−5146.210
S6	91.682513	−41.686207	0.01462753	0.00827200	−5737.395
S7	1635.963518	−443.712607	0.27261055	0.07748191	−5739.888
S8	50.622840	−2.886825	0.01416134	0.00118304	−5487.094
S9	63.159483	−22.926741	0.02151486	0.01012760	−5065.510
S10	76.685731	−5.368403	0.01259531	0.00104879	−5970.702

AICc comparison and selected margins:

Stock	Normal	Student's t	NIG	Selected
S1	−6033.994	−6031.982	−6030.215	Normal
S2	−5512.706	−5627.360	−5622.063	Student's t
S3	−5096.513	−5238.463	−5281.641	NIG
S4	−6382.974	−6380.962	−6379.120	Normal
S5	−5090.895	−5150.869	−5146.210	Student's t
S6	−5547.203	−5688.740	−5737.395	NIG
S7	−5743.694	−5741.682	−5739.888	Normal
S8	−5295.107	−5493.288	−5487.094	Student's t
S9	−4919.506	−5036.762	−5065.510	NIG
S10	−5835.325	−5974.857	−5970.702	Student's t

Normal is selected for S1, S4, and S7. Their fitted t degrees of freedom are extremely large, so Student's t has effectively collapsed to a Normal and its extra parameter is not justified. Student's t is selected for S2, S5, S8, and S10, where finite degrees of freedom capture symmetric heavy tails. NIG is selected for S3, S6, and S9, where its additional skewness and tail flexibility produce the largest AICc improvement.

3(b) Gaussian and Student's t copulas

For each selected margin, I transformed observed returns to uniforms with its fitted CDF. I then fitted each copula by maximum likelihood. For the Gaussian copula, the correlation estimate is the sample correlation matrix of the normal scores $\Phi^{-1}(u)$. For the Student's t copula, I estimated the correlation matrix from the t scores $t_v^{-1}(u)$ at each trial v and maximized the copula log likelihood over v in [2.01, 100]; this profile approach approximates the joint maximum likelihood estimate of the correlation matrix and v. Each copula estimates the same 45 correlations, and the Student's t copula adds v, so I use k = 45 for the Gaussian and k = 46 for the Student's t in AICc and BIC. The Gaussian copula correlation matrix is:

	S1	S2	S3	S4	S5	S6	S7	S8	S9	S10
S1	1.0000	0.5484	0.5344	0.5606	0.5454	0.2938	0.2504	0.2444	0.2712	0.2995
S2	0.5484	1.0000	0.5664	0.5278	0.5188	0.3253	0.2472	0.2529	0.2932	0.2998
S3	0.5344	0.5664	1.0000	0.5315	0.5742	0.3139	0.3093	0.2688	0.3275	0.3083
S4	0.5606	0.5278	0.5315	1.0000	0.5596	0.3035	0.2440	0.2688	0.2816	0.2924
S5	0.5454	0.5188	0.5742	0.5596	1.0000	0.3330	0.2803	0.2977	0.2764	0.2993
S6	0.2938	0.3253	0.3139	0.3035	0.3330	1.0000	0.5455	0.5279	0.5490	0.5628
S7	0.2504	0.2472	0.3093	0.2440	0.2803	0.5455	1.0000	0.5299	0.5406	0.5433
S8	0.2444	0.2529	0.2688	0.2688	0.2977	0.5279	0.5299	1.0000	0.5227	0.5229
S9	0.2712	0.2932	0.3275	0.2816	0.2764	0.5490	0.5406	0.5227	1.0000	0.5423
S10	0.2995	0.2998	0.3083	0.2924	0.2993	0.5628	0.5433	0.5229	0.5423	1.0000

The minimum eigenvalue is 0.37844635, so the matrix is positive definite and requires no repair. The Student's t copula correlation matrix at the fitted v is nearly identical (every entry within 0.001 of the Gaussian matrix, minimum eigenvalue 0.37806972), so the comparison below is driven by the likelihood and the extra parameter v rather than by different dependence estimates.

Copula	Log likelihood	AICc	BIC	v
Gaussian	2113.924	−4133.508	−3916.999	—
Student's t	2111.601	−4126.665	−3905.445	99.999997

The Gaussian copula fits better because it has both the lower AICc and the lower BIC. The fitted t degrees of freedom reached the upper search boundary of approximately 100, which is close to the Gaussian limit. Therefore, the data do not provide enough evidence that an additional common heavy-tail dependence parameter improves the copula. I use the Gaussian copula in part (c).

3(c) Copula simulation, VaR, and ES

I generated 50,000 draws from the fitted Gaussian copula using random seed 545. I transformed each simulated uniform through the inverse CDF of its selected marginal distribution. Each position was valued at its last observed price, and its one-day profit and loss was calculated as position value multiplied by the simulated arithmetic return. For each stock, book, and the total portfolio, 95% absolute VaR is the negative fifth percentile of P&L, and ES is the negative mean P&L in observations at or below that percentile.

Position	Current value	95% VaR	95% ES

Position	Current value	95% VaR	95% ES
S1	$48,329.48	$898.54	$1,132.97
S2	$46,697.43	$1,069.61	$1,556.43
S3	$24,267.28	$803.61	$1,200.53
S4	$43,596.81	$678.47	$855.20
S5	$37,727.45	$1,094.17	$1,521.92
S6	$46,326.64	$1,212.25	$1,846.59
S7	$40,144.31	$878.66	$1,101.91
S8	$26,045.88	$648.89	$1,031.66
S9	$98,248.71	$3,365.61	$5,161.64
S10	$24,512.38	$477.48	$706.66
Book A	**$200,618.44**	**$3,615.35**	**$4,886.64**
Book B	**$235,277.92**	**$5,461.00**	**$7,933.73**
Total portfolio	**$435,896.36**	**$7,640.46**	**$10,648.60**

S9 is the largest stand-alone risk source, with 95% VaR of $3,365.61 and ES of $5,161.64. This reflects both its large $98,248.71 position value and its NIG return margin, which allows skewness and heavier downside behavior. Book B consequently has more risk than Book A: Book B's VaR is $5,461.00 versus $3,615.35 for Book A, and its ES is $7,933.73 versus $4,886.64.

Risk is not additive. The total portfolio VaR of $7,640.46 is 31.34% below the sum of the ten stand-alone stock VaRs. Total ES of $10,648.60 is 33.92% below the sum of the stand-alone ES values. Correlations are generally around 0.52–0.57 within each book but only about 0.24–0.33 across books. Therefore, combining the two books provides meaningful diversification even though all positions are long and all pairwise dependence is positive.

ES exceeds VaR at the stock, book, and portfolio levels because ES averages the more severe losses beyond the VaR threshold. The gap is especially visible for the heavy-tailed or skewed margins, such as S9, and is why ES is useful alongside VaR when evaluating tail severity.